

CTEC 651 – Internet of Things (IoT) Digital Repository Project 1 due 21 September 2026

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Course: CTEC 651 – Internet Technologies Discovery

Due Date: 28 September 2026

Project Title

Building a Local IoT Digital Repository: From Sensor Simulation to Data Visualization

Project Overview

In this project, you will design and implement a complete Internet of Things (IoT) data pipeline using **free and open-source tools**. You will simulate IoT sensor data, transmit it using a messaging protocol, store it in a database, and visualize the results.

This project mirrors real-world IoT systems used in:

- Smart agriculture
 - Healthcare monitoring
 - Smart cities
 - Industrial automation
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Learning Objectives

By completing this project, you will be able to:

- Understand the architecture of IoT systems
 - Simulate real-world sensor data
 - Implement MQTT-based data communication
 - Store and manage time-series data
 - Build data visualizations for insight generation
 - Present a complete end-to-end system
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Tools and Technologies (All Free)

You are required to use only free tools:

- **Python** (data simulation and processing)
 - **MQTT (HiveMQ Public Broker)** – `broker.hivemq.com`
 - **SQLite** (local database)
 - **Matplotlib or Google Sheets** (visualization)
 - *(Optional)* Streamlit (dashboard)
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Project Architecture (Conceptual)

Sensor Simulator → MQTT Broker → Subscriber → Database → Visualization

Project Phases and Deliverables

Phase 1: Project Design & Planning

Objective

Define a real-world IoT use case and design your system.

Examples of Use Cases

- Smart farm temperature monitoring
- Smart home energy tracking
- Patient health monitoring system

Deliverables due 7 September 2026

- 1-page project proposal including:
 - Problem statement
 - Use case description
 - Data fields (e.g., timestamp, temperature, humidity)
 - System architecture diagram
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Phase 2: IoT Data Simulation

Objective

Generate realistic IoT sensor data using Python.

Example Code

```
import random
import time
import json

while True:
    data = {
        "timestamp": time.time(),
        "temperature": round(random.uniform(20, 30), 2),
        "humidity": round(random.uniform(40, 60), 2)
    }
    print(json.dumps(data))
    time.sleep(2)
```

Deliverables due 14 September 2026

- Python script
 - Sample output (at least 10–20 records)
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Phase 3: Data Transmission Using MQTT

Objective

Transmit sensor data using the MQTT protocol.

MQTT Broker

Use the free public broker:

broker.hivemq.com

Example Publisher Code

```
import paho.mqtt.client as mqtt
import json, random, time

client = mqtt.Client()
client.connect("broker.hivemq.com", 1883, 60)
```

```
while True:
    data = {"temperature": random.uniform(20,30)}
    client.publish("ctec651/sensor", json.dumps(data))
    time.sleep(2)
```

Example Subscriber Code

```
def on_message(client, userdata, msg):
    print(msg.payload.decode())

client = mqtt.Client()
client.connect("broker.hivemq.com", 1883, 60)
client.subscribe("ctec651/sensor")
client.on_message = on_message
client.loop_forever()
```

Deliverables due 21 September 2026

- Publisher and subscriber scripts
 - Screenshot or output showing data transmission
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Phase 4: Data Storage

Objective

Store IoT data in a local database.

Database

Use SQLite (built into Python)

Example Code

```
import sqlite3

conn = sqlite3.connect("iot_data.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS sensor_data (
    timestamp REAL,
    temperature REAL
)
""")
```

Deliverables due 28 September 2026

- SQLite database file
 - Table schema
 - At least 50 stored records
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Phase 5: Data Visualization

Objective

Visualize IoT data to identify trends.

Option A: Python Visualization

```
import matplotlib.pyplot as plt
import sqlite3

conn = sqlite3.connect("iot_data.db")
data = conn.execute("SELECT * FROM sensor_data").fetchall()

timestamps = [row[0] for row in data]
temps = [row[1] for row in data]

plt.plot(timestamps, temps)
plt.title("Temperature Over Time")
plt.xlabel("Time")
plt.ylabel("Temperature")
plt.show()
```

Option B: Google Sheets / Excel

- Export data as CSV
- Create charts

Deliverables due 28 September 2026

- At least **two visualizations**
 - Short written interpretation (What does the data show?)
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Phase 6 (Optional – Bonus): Dashboard

Objective

Create a simple dashboard interface.

Tool

Streamlit (optional)

Deliverables

- Simple dashboard OR integrated script
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Phase 7: Final Presentation due 28 September 2026

Objective

Present your complete IoT system.

Presentation Must Include

- Problem and use case
- System architecture
- Live demo (or recorded demo)
- Key insights from your data

Deliverables due 28 September 2026

- Slides (8–15 slides recommended)
 - Demo
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Grading Breakdown

Component	Weight
Phase 1: Design & Planning	15%
Phases 2–3: Data Simulation & Transmission	20%
Phase 4: Data Storage	20%
Phase 5: Visualization	20%
Final Presentation	25%

Submission Requirements

Submit the following:

- Python scripts
 - Database file
 - Visualizations (images or files)
 - ~~Project report (optional but recommended)~~
 - Presentation slides
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Important Notes

- Use only **free tools** (no paid services required)
 - Ensure your project runs **without errors**
 - Stay organized—each phase builds on the previous one
 - **Creativity is encouraged** (custom sensors, additional features)
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Next Steps (Optional Enhancements - Bonus)

Students who want to go further may:

- Add anomaly detection
 - Use multiple sensors
 - Build a web dashboard
 - Integrate basic AI models
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